Speechmaker 6000:

Plan:
Why is our problem important:
There is real demand for space mining-
AstroForge is targeting metallic asteroids for precious metals, while Karman+ is going after hydrated asteroids to extract water. Source: The Space Review. thespacereview
TransAstra is running a study to capture a small near-Earth asteroid for use in space industries. Wboc.
SpaceX included asteroid mining as one of the future markets it may pursue in its IPO prospectus. Isstracker.

There is a large demand for further space exploration, however this requires many materials, which are very costly to export from earth. If space mining was used, then space projects are able to travel further from earth, and when they need a top up of materials, then they can simply get some materials from the space mining.

Currently, we can determine which asteroids contain certain materials, however, the issue is determining which ones to mine relating to which products you want to create and where do we build the stations to process them. If we want to create product A, some of the materials required for it may be sparsely located in space.

Data: 65,000 real asteroids from IMCCE's SsODNet database, each with a size and composition class.
Resources: Composition class → minerals → tonnes of water, iron, nickel, aluminium and so on, based on meteorite research.
Hubs: K-means places each station at the weighted centre of the asteroids it serves, so the more valuable an asteroid, the harder it pulls its station towards it.
Placement: Station locations are refined to minimise transport cost, accounting for whether ore is refined at the mine or shipped raw.

Show off the project!

Limitations: position is class-level, positions are a snapshot, and distance stands in for fuel cost. Then turn each into next steps: orbital mechanics, delta-v costs and real mission data. Judges respect this, and it pre-empts the hardest questions.

Close off on the main benefit e.g. End on the bootstrapping idea, framing the project as the planning tool for a self-sustaining space industry. Every station we place makes the next one cheaper. This is how a space economy starts.

Potential questions to answer:

"Why ignore orbits?"- Many asteroids fixed position relative to the sun, This is a good initial solution to begin space mining, and orbits can be accounted for. As there are

"How accurate is the composition data?", "Orbits are precise and sizes are good to about 10%, but compositions are class averages from mostly colour-based classifications, so our resource estimates are reliable as comparisons between regions and types, not as exact tonnages for individual asteroids." See claude

“Why k mean and not something else?”



Things to say during presentation:

20 secs hook:
Everything we build in space today is launched from Earth, costing thousands of pounds per kilogram. The materials we need are already out there, in asteroids.

25 secs problem statement:
Knowing which asteroids are valuable isn’t enough. No single asteroid has everything you need, so the real question is a logistical one, which asteroids do we mine, and where do we build the stations that turn raw rock into useful products? Imagine if you were in Medievel times, and you were the first person to receive a map of all the resources in the local area. We provide the map so that space agencies can build vast networks of celestial mines and supply much deeper travel into our solar system.

60 secs how does our project work:
Data: 65,000 real asteroids from IMCCE’s SsODNet database, each with a size and composition class.

Resources: Composition class → minerals → tonnes of water, iron, nickel, aluminium and so on, based on meteorite research.

DAG: ore→ raw materials → product. john u got this

Mining location: . K-means places each station at the weighted centre of the asteroids it serves, so the more valuable an asteroid, the harder it pulls its station towards it..

Monte carlo: Randomly sample class types to determine the range of each class. Many runs average it out.

Placement: Manufacturing locations are refined to minimise transport cost from the mines.

30 secs (applications):

Station one produces the propellant and metal needed to build station two. This means that a self sustaining network of these mines can be creating without ever needing to receive resources from earth. (Bootstrapping)

Limitations 30 seconds match every limitation with a future solution:

not enough asteroids. Solution larger data set
DAG abstracted. Solution More throrough DAG
better model for the amount of materials which can be gathered by class. Solution
Data is limited based on what we can observe from earth. Solution- it will be accurate for a large group of asteroids.
positions are a snapshot. Solution delta-v costs calculate how much spaceships need to change velocity to maneuvere around an object.
distance stands in for fuel cost. Solution real mission data.

15 secs closing argument:
End on the bootstrapping idea, framing the project as the planning tool for a self-sustaining space industry.

Things to go on website:

"We plan to pursue asteroid mining operations to extract metals and other critical resources from near-Earth and main-belt asteroids, providing abundant raw materials for space-based industries and reducing the need to launch mass from Earth," SpaceX stated in their prospectus.

Can u make another mine from the asteroids can u make fuel from the asteroids